

Factor Validation: Commodity Strategies

26 commodities · monthly 1980-01–2026-01 · source: GitHub mirror of the World Bank file; Updated on February 03, 2026 · generated 2026-09-27

Headline numbers

VALUE (5Y) NET SHARPE

0.95

t = 5.5 (Newey-West)

TSMOM (12-1) NET SHARPE

0.40

t = 1.9

XS MOMENTUM NET SHARPE

−0.28

t = −1.6

PBO, MOMENTUM GRID

37%

chance the TSMOM winner is below median OOS

VARIANTS TRIED

46

every one is charged in the deflated Sharpe

LAG-1 AUTOCORRELATION

0.26

vs 0.01 at lag 2: averaging artefact

- **Value is the standout and it survives every check.** A 5-year reversal signal earns a net Sharpe of 0.95 with a Newey-West t of 5.5 (above the Harvey–Liu–Zhu t > 3 bar), a deflated Sharpe of 1.000, and a post-publication Sharpe of 0.96.
- **Time-series momentum is real but modest.** Net Sharpe 0.40 (0.43 in the paper's window, 0.34 after publication); t = 1.94 falls short of t > 3, and choosing its best look-back has a 37% probability of being overfit.
- **Cross-sectional momentum does not replicate on spot prices** (net Sharpe −0.28; 0 of its 20 variants is positive). The paper used futures returns; see the caveat below.
- **Value and momentum hedge each other** (correlation −0.54 with XS momentum, −0.35 with TSMOM), as Asness, Moskowitz and Pedersen report across asset classes.
- **Parameter selection is stable.** Out-of-sample Sharpe from purged CPCV is within 0.11 of the in-sample best in every family: rankings barely change between halves of the sample, which is what a low PBO means.

Caveat. Spot is not futures. The Pink Sheet reports monthly *average spot* prices. Spot prices mean-revert more than futures returns do, because futures curves already price expected reversion, so value's Sharpe here is an upper bound for a tradable version. Monthly averaging also smooths volatility and adds lag-1 autocorrelation, which is why every signal skips the most recent month.

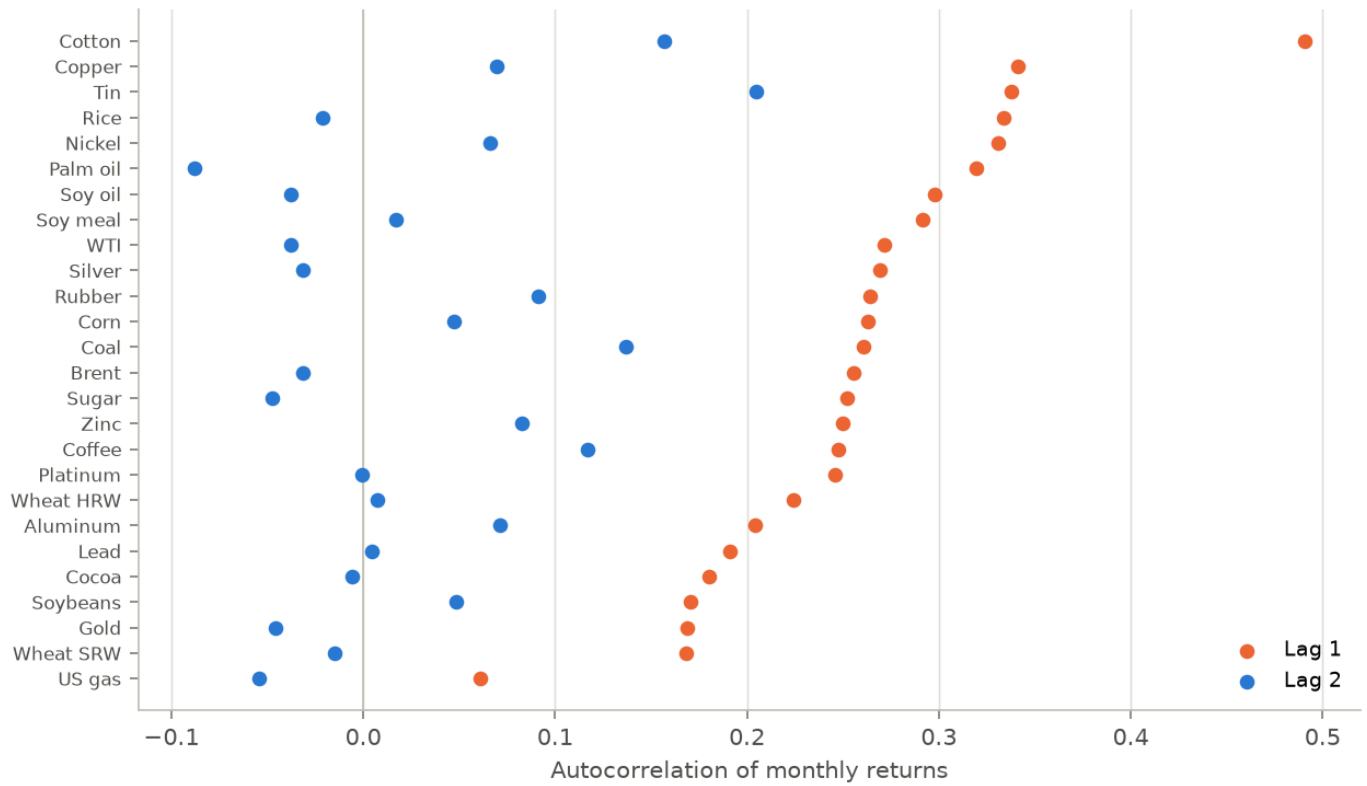
1. Data

World Bank *Pink Sheet* monthly prices for 26 commodities with a liquid futures analogue, 1980-01 to 2026-01. Returns are monthly log changes. A month is tradable once at least 10 commodities have prices.

Universe

Commodity	Sector	Unit	Ann. vol	Lag-1 autocorr	Lag-2 autocorr
Brent	Energy	(\$/bbl)	31.3%	0.26	-0.03
WTI	Energy	(\$/bbl)	32.0%	0.27	-0.04
US gas	Energy	(\$/mmbtu)	46.9%	0.06	-0.05
Coal	Energy	(\$/mt)	24.1%	0.26	0.14
Aluminum	Base metals	(\$/mt)	18.7%	0.20	0.07
Copper	Base metals	(\$/mt)	21.0%	0.34	0.07
Lead	Base metals	(\$/mt)	22.8%	0.19	0.00
Nickel	Base metals	(\$/mt)	28.2%	0.33	0.07
Tin	Base metals	(\$/mt)	20.2%	0.34	0.20
Zinc	Base metals	(\$/mt)	21.5%	0.25	0.08
Gold	Precious	(\$/troy oz)	13.9%	0.17	-0.05
Silver	Precious	(\$/troy oz)	26.1%	0.27	-0.03
Platinum	Precious	(\$/troy oz)	20.4%	0.25	-0.00
Corn	Grains	(\$/mt)	20.3%	0.26	0.05
Wheat SRW	Grains	(\$/mt)	22.6%	0.17	-0.02
Wheat HRW	Grains	(\$/mt)	19.9%	0.22	0.01
Soybeans	Grains	(\$/mt)	17.8%	0.17	0.05
Soy oil	Grains	(\$/mt)	20.1%	0.30	-0.04
Soy meal	Grains	(\$/mt)	18.9%	0.29	0.02
Rice	Grains	(\$/mt)	19.4%	0.33	-0.02
Palm oil	Grains	(\$/mt)	25.0%	0.32	-0.09
Cocoa	Softs	(\$/kg)	22.4%	0.18	-0.01
Coffee	Softs	(\$/kg)	24.7%	0.25	0.12
Sugar	Softs	(\$/kg)	29.8%	0.25	-0.05
Cotton	Softs	(\$/kg)	18.4%	0.49	0.16
Rubber	Softs	(\$/kg)	23.7%	0.26	0.09

Monthly averaging shows up as lag-1 autocorrelation



Averaging within the month creates lag-1 autocorrelation (≈ 0.25 for a random walk, Working 1960); lag 2 is clean.

2. Replicating the published strategies

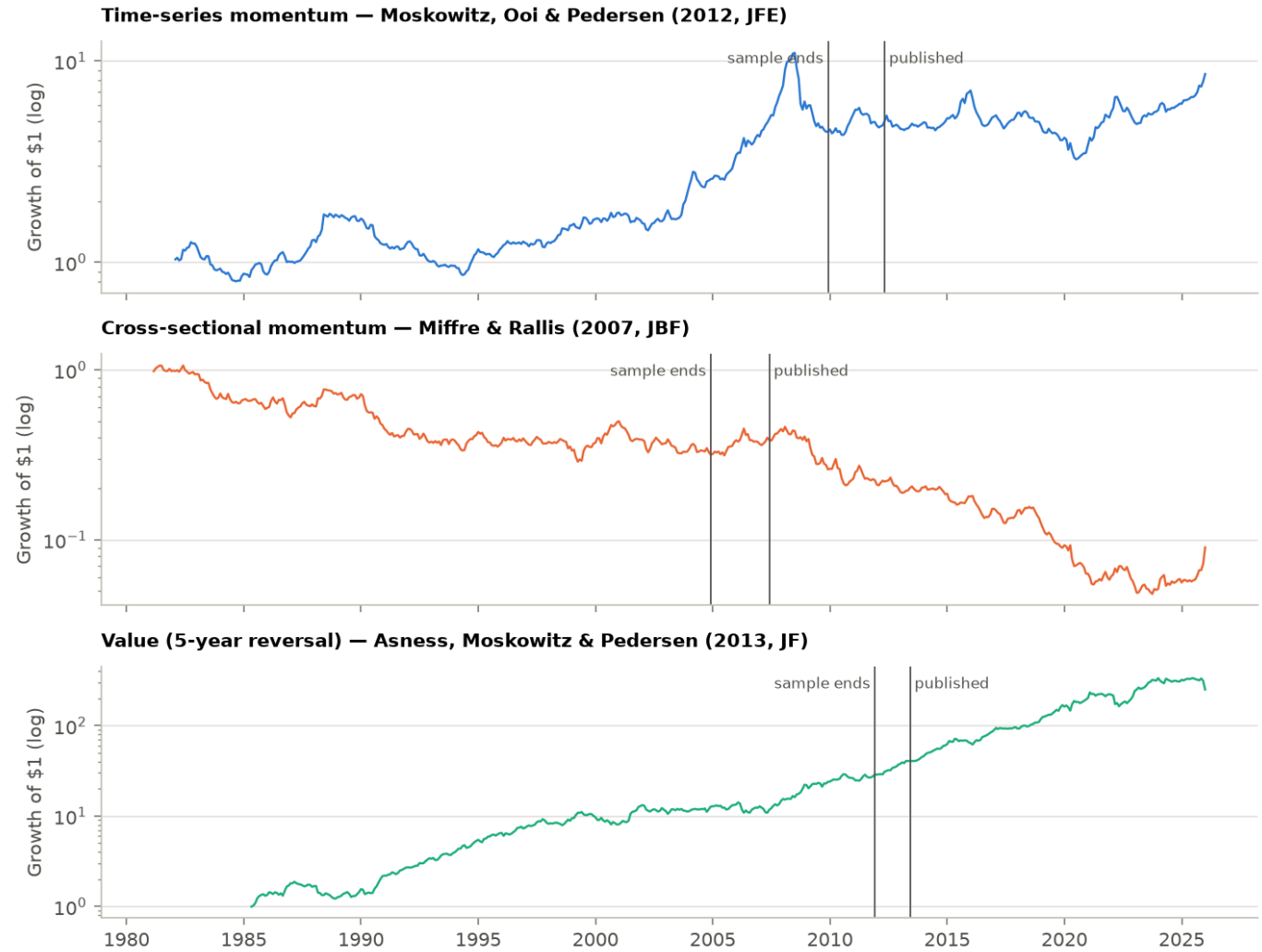
What is being replicated

Strategy	Paper	Settings	Sample ends	Published
Time-series momentum	Moskowitz, Ooi & Pedersen (2012, JFE)	lookback=12, skip=1, scaled=True	2009-12	2012-05
Cross-sectional momentum	Miffre & Rallis (2007, JBF)	lookback=12, skip=1, q=0.333	2004-12	2007-06
Value (5-year reversal)	Asness, Moskowitz & Pedersen (2013, JF)	years=5, q=0.333	2011-12	2013-06

Full-sample performance (costs: 10 bp per unit of turnover)

Strategy	Returns	Months	Ann. return	Ann. vol	Sharpe	NW t	Max DD	Turnover/mo
TSMOM	gross	528	6.6%	14.9%	0.44	2.13	-68%	0.48
TSMOM	net	528	6.0%	14.9%	0.40	1.94	-70%	0.48
XSMOM	gross	539	-3.4%	15.1%	-0.22	-1.29	-94%	0.71
XSMOM	net	539	-4.2%	15.1%	-0.28	-1.62	-95%	0.71
VALUE	gross	489	15.3%	15.6%	0.98	5.68	-35%	0.41
VALUE	net	489	14.8%	15.6%	0.95	5.51	-35%	0.41
COMBO	net	489	5.6%	7.4%	0.75	4.47	-18%	–

Published strategies, net of costs

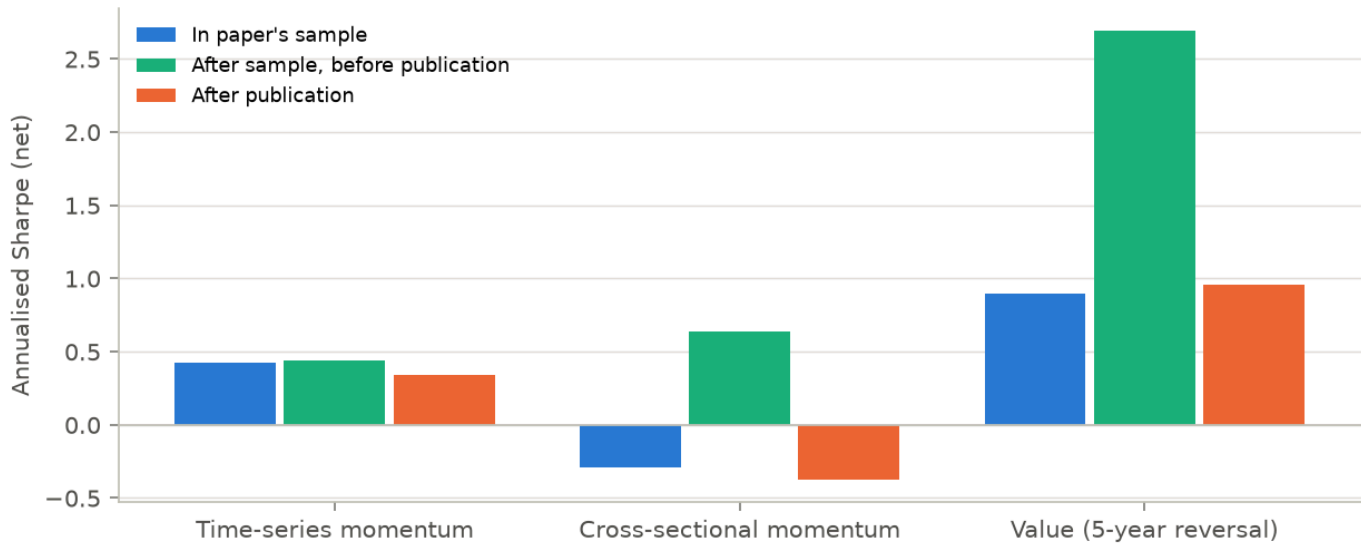


Growth of \$1, net of costs, with each paper's sample end and publication date.

McLean-Pontiff windows: before and after the paper

Strategy	Window	From	To	Months	Sharpe	NW t
TSMOM	In paper's sample	1982-02	2009-12	335	0.43	1.60
TSMOM	After sample, before publication	2010-01	2012-05	29	0.44	0.55
TSMOM	After publication	2012-06	2026-01	164	0.34	1.01
XSMOM	In paper's sample	1981-03	2004-12	286	-0.29	-1.29
XSMOM	After sample, before publication	2005-01	2007-06	30	0.64	1.07
XSMOM	After publication	2007-07	2026-01	223	-0.38	-1.36
VALUE	In paper's sample	1985-05	2011-12	320	0.90	4.13
VALUE	After sample, before publication	2012-01	2013-06	18	2.69	4.47
VALUE	After publication	2013-07	2026-01	151	0.96	3.23

Does the edge survive publication?



Sharpe ratio in each window. Short windows (under 3 years) are noisy.

Placebo test. Shuffling each month's positions across commodities keeps the portfolio's shape but destroys which commodity gets which position. For the market-neutral strategies the placebos centre near zero (XS momentum -0.00 , value 0.02), so the portfolio code carries no structural bias, and value sits far in the placebo tail ($p = 0.005$). TSMOM is different: a shuffle keeps each month's net long/short tilt, and that alone earns a Sharpe of 0.38 against the real 0.44 . Most of time-series momentum's edge here is timing the commodity complex as a whole, not picking commodities.

Permutation test (200 shuffles, gross of costs)

Strategy	Real gross Sharpe	Placebo mean	Placebo 5–95%	p-value
TSMOM	0.44	0.38	0.22 to 0.53	0.239
XSMOM	-0.22	-0.00	-0.22 to 0.21	0.945
VALUE	0.98	0.02	-0.21 to 0.25	0.005

Link to the regime workflow. Months are labelled by the WTI regime the Markov-switching model reported the month before (so the label is known in advance). Turbulent months are few, so treat this as descriptive.

Strategy performance by energy regime (2007 onward)

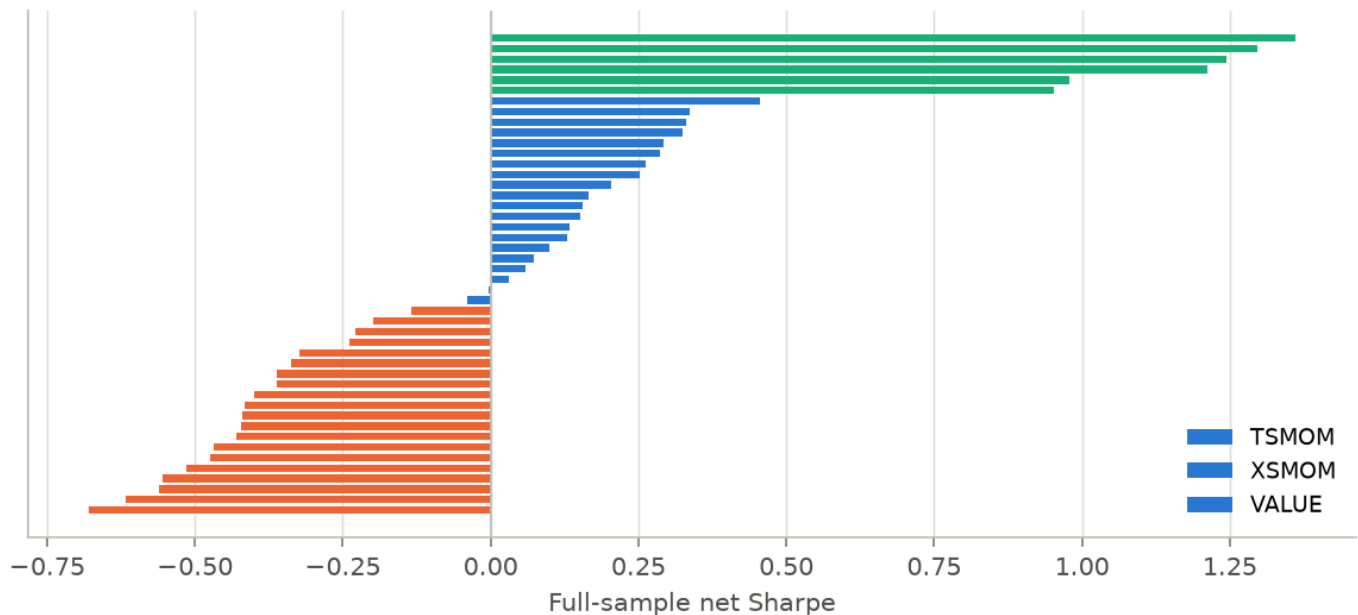
Strategy	WTI regime last month	Months	Ann. return	Sharpe
TSMOM	Calm	210	9.9%	0.65
TSMOM	Turbulent	18	-48.7%	-1.71
XSMOM	Calm	210	-3.1%	-0.19
XSMOM	Turbulent	18	-38.9%	-1.67
VALUE	Calm	210	15.4%	1.04
VALUE	Turbulent	18	35.9%	1.43

3. Choosing parameters without fooling yourself

The research grid holds **46 variants**: look-backs of 1–12 months, one- or two-month skips, vol-scaling on or off, terciles or quartiles, and 3–5 year value windows. Picking the best one after seeing the results is the classic way to overfit.

Combinatorial purged cross-validation (López de Prado, 2018) splits 1980–2026 into 10 blocks, holds out 2 at a time (45 splits, 9 complete out-of-sample paths), chooses the best variant on the rest, and drops 1 month before and 12 months after each held-out block so overlapping look-back windows cannot leak.

All 46 variants in the research grid

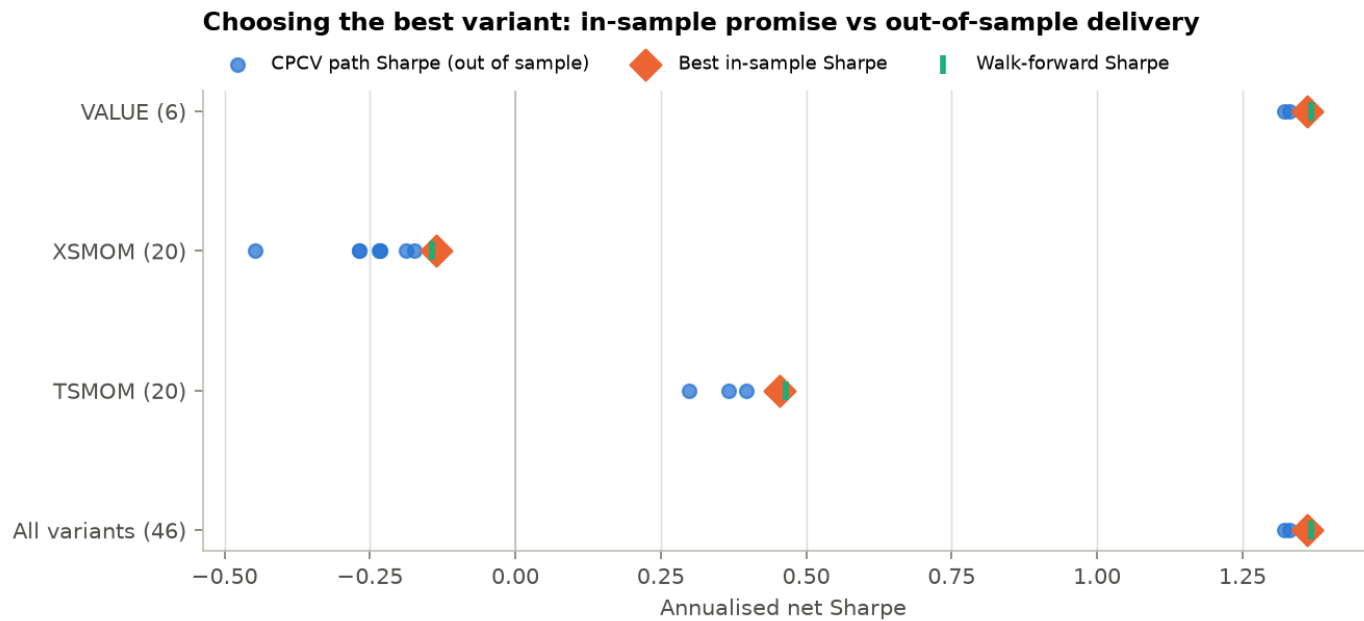


Full-sample net Sharpe of every variant, coloured by family.

In-sample promise vs out-of-sample delivery (annualised net Sharpe)

Variant set	N	Method	In-sample best	Out-of-sample	Shortfall	Most chosen
All variants	46	Purged CPCV	1.36	1.35	0.01	VALUE 4y q=1/4
All variants	46	Naive CV	1.36	1.36	0.00	VALUE 4y q=1/4
All variants	46	Walk-forward	1.36	1.37	-0.01	VALUE 3y q=1/3
TSMOM	20	Purged CPCV	0.45	0.42	0.03	TSMOM 12-1 vol-scaled
TSMOM	20	Naive CV	0.45	0.45	-0.00	TSMOM 12-1 vol-scaled
TSMOM	20	Walk-forward	0.45	0.47	-0.01	TSMOM 12-1 vol-scaled
XSMOM	20	Purged CPCV	-0.14	-0.24	0.11	XSMOM 9-2 q=1/4
XSMOM	20	Naive CV	-0.14	-0.23	0.10	XSMOM 9-2 q=1/4
XSMOM	20	Walk-forward	-0.14	-0.14	0.01	XSMOM 9-2 q=1/4
VALUE	6	Purged CPCV	1.36	1.35	0.01	VALUE 4y q=1/4
VALUE	6	Naive CV	1.36	1.36	0.00	VALUE 4y q=1/4

Variant set	N	Method	In-sample best	Out-of-sample	Shortfall	Most chosen
VALUE	6	Walk-forward	1.36	1.37	-0.01	VALUE 3y q=1/3



Each dot is one CPCV path. The diamond is what a backtest of the best variant would report.

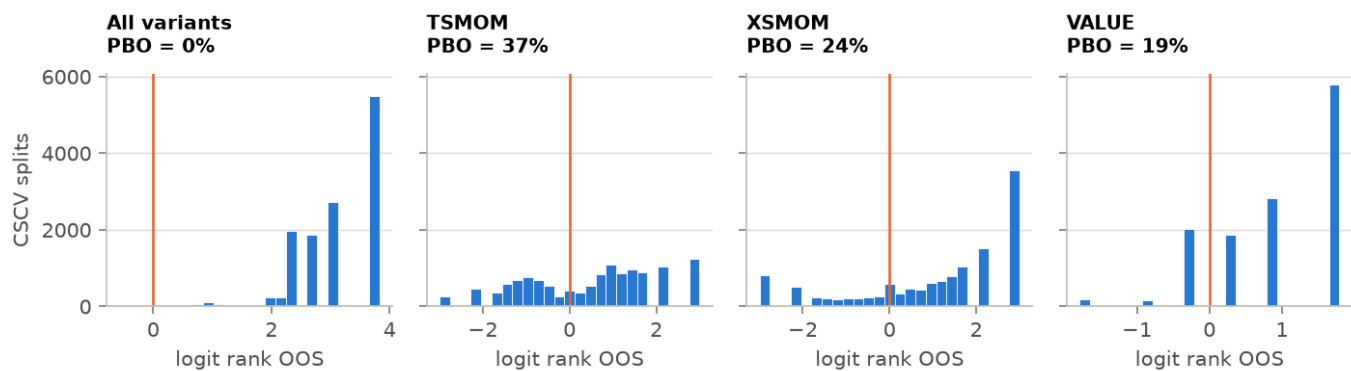
Purging changes little here because the strategies are simple rules: the only thing fitted is the choice of variant, and rankings are stable. Purging matters far more for fitted models (regressions, trees) trained on overlapping labels; the code handles both.

4. Overfitting diagnostics

Probability of backtest overfitting (CSCV, 16 blocks)

Variant set	Variants	CSCV splits	PBO	OOS Sharpe < 0	Verdict
All variants	46	12,870	0.1%	2%	✓ Low risk
TSMOM	20	12,870	37.2%	20%	✓ Low risk
XSMOM	20	12,870	24.5%	81%	✓ Low risk
VALUE	6	12,870	18.5%	0%	✓ Low risk

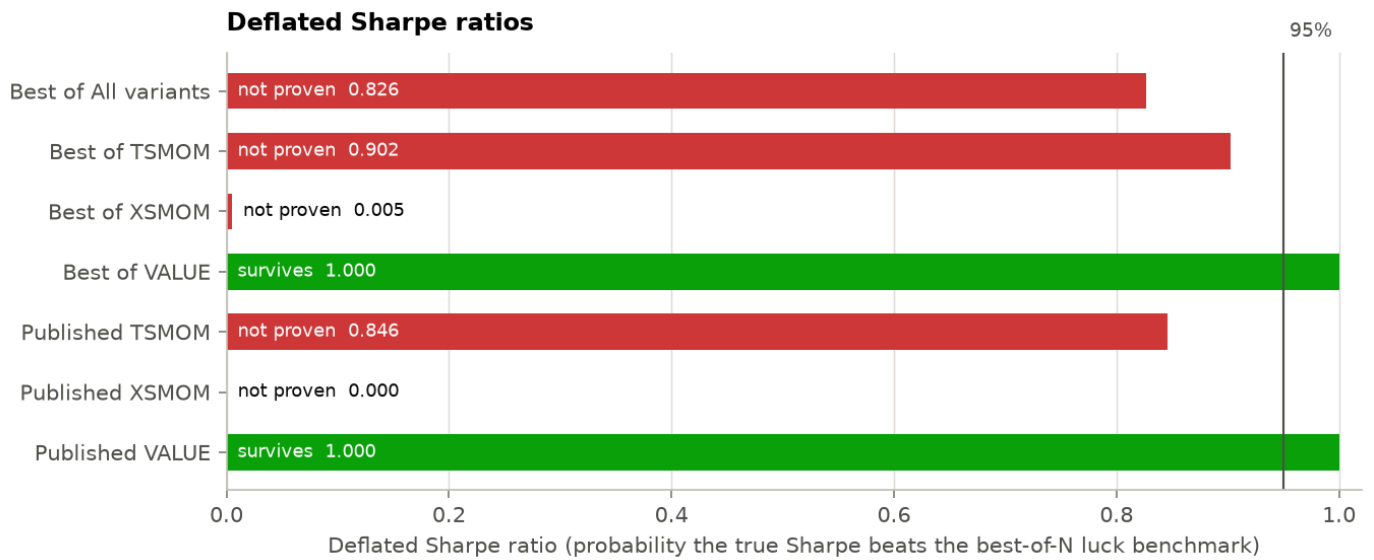
Where does the in-sample winner rank out of sample? (left of red = bottom half)



Logit of the in-sample winner's out-of-sample rank. Mass left of zero = overfitting.

Probabilistic and deflated Sharpe ratios (annualised Sharpe; DSR > 0.95 to pass)

Candidate	Variant	Trials N	Sharpe	Luck bar SR0	PSR vs 0	DSR	NW t	Verdict
Best of All variants	VALUE 4y q=1/4	46	1.36	1.20	1.000	0.826	7.37	✗ Not proven
Best of TSMOM	TSMOM 12-1 vol-scaled	20	0.45	0.25	0.998	0.902	2.10	✗ Not proven
Best of XSMOM	XSMOM 9-2 q=1/4	20	-0.14	0.27	0.194	0.005	-0.78	✗ Not proven
Best of VALUE	VALUE 4y q=1/4	6	1.36	0.22	1.000	1.000	7.37	✓ Survives
Published TSMOM	TSMOM 12-1 vol-scaled	20	0.40	0.25	0.996	0.846	1.94	✗ Not proven
Published XSMOM	XSMOM 12-1 q=1/3	20	-0.28	0.27	0.032	0.000	-1.62	✗ Not proven
Published VALUE	VALUE 5y q=1/3	6	0.95	0.22	1.000	1.000	5.51	✓ Survives



Deflated Sharpe ratios; bars in green pass the 95% bar.

Reading the luck bar. SR0 is the Sharpe the best of N variants would show with zero true skill, given how much the variants' Sharpes differ. Across all 46 variants the bar is high because the families differ in genuine skill, not only luck; within a family it is the fairer test, which is why each published strategy is deflated by its own family.

5. Validation summary

Validation evidence

Check	How	Result	Status
Look-ahead	Signals use prices through $t-1$; placebo and no-look-ahead tests	Market-neutral placebos centre on 0.02 or less	✓ Pass
Data artefact	Lag-1 autocorrelation from monthly averaging	Median 0.26; neutralised by skipping a month	✓ Pass
Replication	Published settings, gross and net	Positive net Sharpe: TSMOM, VALUE; negative: XSMOM	▲ Partial
Decay	McLean–Pontiff windows	Post-publication Sharpe: value 0.96, TSMOM 0.34	✓ Pass
Selection bias	Purged CPCV, walk-forward	Out-of-sample within 0.11 of in-sample in every family	✓ Pass
Overfitting	PBO (CSCV), deflated Sharpe	PBO 37% for TSMOM; value DSR 1.000	▲ Family-dependent

Limitations

- Spot, monthly-average prices: not the futures excess returns the papers use. No roll yield, so no carry strategy.
- Costs are a flat 10 bp per unit of turnover; real costs vary by market and by era.
- The grid is what one researcher might try. The literature as a whole tried far more, so true N is larger.
- The regime split has only 18 turbulent months: descriptive only.

References

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